Summary in Graph

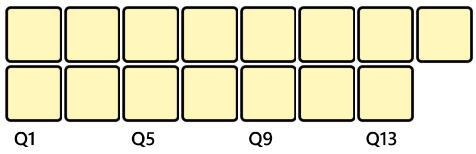
Exam Summary (GO Classes DA Test Series 2026 | Linear Algebra | Topic Wise Test 4)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (43) out of 432 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The closest point to $\begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$ in the subspace of $\mathbb{R}^3$ spanned by $\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix}$ is

A. $\begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix}$

B. $\begin{bmatrix} 3 \\ 1/2 \\ 0 \end{bmatrix}$

C. $\begin{bmatrix} 3 \\ 0 \\ 5 \end{bmatrix}$

D. $\begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following conditions must be satisfied non zero by the vectors $\mathbf{a}_1, \mathbf{a}_2 \in \mathbb{R}^m$ for the matrix

$$\frac{\mathbf{a}_1\mathbf{a}_1^T}{\mathbf{a}_1^T\mathbf{a}_1} + \frac{\mathbf{a}_2\mathbf{a}_2^T}{\mathbf{a}_2^T\mathbf{a}_2}$$

to be the projection matrix onto the span of $\mathbf{a}_1$ and $\mathbf{a}_2$?

A. $\mathbf{a}_1$ and $\mathbf{a}_2$ are independent.
B. $\mathbf{a}_1$ and $\mathbf{a}_2$ are orthogonal.
C. $\mathbf{a}_1$ and $\mathbf{a}_2$ are parallel.
D. $\mathbf{a}_1$ and $\mathbf{a}_2$ are orthonormal.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let Q be a 5×3 matrix with orthonormal columns.
Which of the following statements must be true?

A. $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^3$
B. $\|Q^Ty\| = \|y\|$ for all $y \in \mathbb{R}^5$
C. Both A and B are true.
D. Neither A nor B is true.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Find the projection of the vector $(1, -1, 0)^T$ onto the plane $x + y + z = 0$ in $\mathbb{R}^3$.

A. $(1, -1, 2)^T$
B. $(2, 1, -3)^T$
C. $(1, -1, 0)^T$
D. $(2, -2, 0)^T$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose that $\mathbf{x}, \mathbf{y}$, and $\mathbf{z}$ are independent vectors in $\mathbf{R}^3$. Moreover, assume that $\mathbf{x}$ and $\mathbf{y}$ are orthonormal (i.e., $\|\mathbf{x}\| = 1, \|\mathbf{y}\| = 1$, and $\mathbf{x} \cdot \mathbf{y} = 0$).
What is the linear combination of $\mathbf{x}$ and $\mathbf{y}$ that is closest to $\mathbf{z}$?

A. $(\mathbf{z} \cdot \mathbf{x})\mathbf{x} + (\mathbf{z} \cdot \mathbf{y})\mathbf{y}$
B. $(\mathbf{z} \cdot \mathbf{x})\mathbf{y} + (\mathbf{z} \cdot \mathbf{y})\mathbf{x}$
C. $\|\mathbf{z}\|\mathbf{x} + \|\mathbf{z}\|\mathbf{y}$
D. $(\mathbf{z} \cdot \mathbf{x})\mathbf{x} - (\mathbf{z} \cdot \mathbf{y})\mathbf{y}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the vector $\vec{w} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ and the line $W = \text{Span}\{\vec{w}\}$ spanned by $\vec{w}$. $A\vec{x}$ computes the projection of a vector $\vec{x}$ onto W if

A. $A = \begin{bmatrix} 1/10 & -3 \\ -3/10 & 9 \end{bmatrix}$
B. $A = \begin{bmatrix} 9/10 & -3/10 \\ -3/10 & 1/10 \end{bmatrix}$
C. $A = \begin{bmatrix} 1/10 & -3/10 \\ -3/10 & 1/10 \end{bmatrix}$
D. $A = \begin{bmatrix} 1/10 & -3/10 \\ -3/10 & 9/10 \end{bmatrix}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $W = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\}$. Compute $\text{proj}_W \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

A. $\frac{1}{2}\begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix}$
B. $\begin{bmatrix} 2 \\ 0 \\ 0 \\ -2 \end{bmatrix}$
C. $\frac{1}{2}\begin{bmatrix} -3 \\ -1 \\ 1 \\ 3 \end{bmatrix}$
D. $\begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the orthogonal vectors $\vec{w}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ and $\vec{w}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. The distance from the vector $\vec{x} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$ to $W = \text{Span}\{\vec{w}_1, \vec{w}_2\}$ is:

A. $\sqrt{6}$
B. 1
C. $\sqrt{2}$
D. $\sqrt{3}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Consider the matrices

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 & 11 \\ 3 & 6 & -3 & 1 & 15 \\ -1 & -2 & 1 & 2 & 2 \\ 4 & 8 & -4 & 4 & 28 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where B is the reduced echelon form of A. A basis for the orthogonal complement of the row space of A is given by

A. $\left\{ \begin{bmatrix} 2 \\ 3 \\ -1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix} \right\}$

B. $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 0 \\ -3 \\ 1 \end{bmatrix} \right\}$

C. $\left\{ \begin{bmatrix} -1 \\ -2 \\ 1 \\ 0 \\ -4 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \\ -3 \end{bmatrix} \right\}$

D. $\left\{ \begin{bmatrix} -1 \\ -2 \\ 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ -4 \\ 4 \\ 28 \end{bmatrix} \right\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

If $\boldsymbol{q}_1, \boldsymbol{q}_2, \boldsymbol{q}_3$ form an orthonormal basis for $\mathbb{R}^3$, and T is the transformation that projects every vector $\boldsymbol{v}$ onto the plane of $\boldsymbol{q}_1$ and $\boldsymbol{q}_2$, what is the matrix for T in this basis?

A. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

D. $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

If P projects onto $C\left(A^T\right)$, the row space of some $m \times n$ matrix A, then $(I-P)^2 x$ for any $x \in \mathbb{R}^n$ gives a vector in which fundamental subspace?

A. $C\left(A^T\right)$
B. $N(A)$
C. $C(A)$
D. $N\left(A^T\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider a 10×3 matrix A with an SVD $A=U \Sigma V^T$, where $\Sigma=\begin{pmatrix} 100 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.
Which of the following statements are true regarding the projection matrices?

A. The projection matrix onto the column space $C(A)$ is UU^T.
B. The projection matrix onto the column space $C\left(A^T\right)$ is VV^T.
C. The projection matrix onto the column space $C(A)$ is I (the identity matrix).
D. The projection matrix onto the column space $C\left(A^T\right)$ is I (the identity matrix).

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 04sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Given that A is a symmetric matrix and P is the projection matrix onto $N(A)$, what is PA ?

A. A
B. P
C. I (the identity matrix)
D. 0 (the zero matrix)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Consider the subspace W of $\mathbf{R}^3$ defined by $W = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbf{R}^3 \middle| \; x_1 - x_2 - 2x_3 = 0 \right\}$.

Let B be the matrix representing the orthogonal projection onto W.

Which of the following statements are true?

A. The eigenvalues of B are $\lambda = 0$ and $\lambda = 1$.

B. $B \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

C. The null space of B is 1 -dimensional.

D. The eigenvalues of B are $\lambda = 0$ and $\lambda = -1$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following methods correctly determines the matrix P_V for orthogonal projection onto the plane V defined by the equation $x + 2y - z = 0$?

A. Method A: Find a basis for V, put your basis vectors into a matrix A, and use the formula $P_V = A\left(A^T A\right)^{-1} A^T$.

B. Method B: Compute the matrix for orthogonal projection $P_{V^\perp}$ onto the line $V^\perp$ using the formula $\frac{vv^T}{v \cdot v}$, where $v = (1, 2, -1)$, and subtract: $P_V = I_3 - P_{V^\perp}$.

C. Method C : Find an orthonormal basis for V, put your basis vectors into a matrix Q, and use the formula $P_V = QQ^T$.

D. None of the above methods.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

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